

Peer Review Report

Journal: AI & SOCIETY (Springer) **Topical Collection:** *Responsible use of AI in Art Creation and Archival Practice* (Guest Editors: Farina, Webb, Stahl; Collection Advisor: Stapleton) **Manuscript title:** Disentangling Algorithmic Bias from Archival Artifacts: A Controlled Audit of Vision-Language Model Valuation in Museum Collections
Contribution type assumed: Research Paper (max 10,000 words) **Review type:** Double-blind

Recommendation

MAJOR REVISION — with the explicit caveat that if Major Concern 1 (directional contradiction) cannot be resolved without overturning the paper's headline claims, the appropriate outcome is *Reject and Resubmit* as a new submission.

I want to be clear about why this is not a straightforward "revise and improve." The manuscript is well-organised, the methodological instinct is sound, and the core research programme — separating archival confounding from model behaviour — is genuinely worth publishing. But the paper currently reports results whose **direction contradicts the interpretation placed on them**, in the abstract, in the discussion, in the conclusion, and in the appendix. That is not a presentational problem. Until it is fixed, I cannot assess whether the paper supports its own thesis.

1. Summary of the Submission

The authors harvest 1,500 public-domain records from two Metropolitan Museum of Art departments, infer artist gender from names using (`gender-guesser`), and construct a stratified evaluation cohort of 96 artworks (68 male-attributed, 28 female-attributed). They score each image zero-shot under two ViT-B/32 CLIP variants (OpenAI/WIT and OpenCLIP/LAION-2B) using a softmax probability differential across three high/low value prompt pairs. They test group differences non-parametrically (Mann-Whitney U, rank-biserial r , bootstrap CIs) and then fit OLS regressions adjusting for medium category and image aspect ratio.

The headline claims are: (a) apparent gender disparities under OpenAI CLIP dissolve once medium is controlled; (b) OpenCLIP retains a significant gender effect after adjustment; (c) therefore pretraining curation determines whether bias is metadata-mediated or direct.

2. Fit to the Topical Collection

Partial fit; requires reframing.

The collection explicitly asks contributors to move *beyond* GenAI toward "more longstanding AI techniques," to address "intelligent/smart metadata technologies to

manage cultural metadata," and to explore how "the use of archives might impact the development and efficiency of AI itself." A CLIP audit on museum metadata sits squarely inside aims 3, 4, and 6. The paper is on-topic.

However, AI & SOCIETY is *Knowledge, Culture and Communication*, and the collection is explicitly interdisciplinary — it names sociology, philosophy, law, art performance, and archival studies as target contributor communities. As written, this manuscript reads as a computer-vision fairness paper that has been lightly dressed for a humanities venue. Specifically:

- Critical archival theory appears as a single citation to Bowker & Star, deployed decoratively rather than analytically. The paper asserts that classification systems are "sociotechnical infrastructures" and then never uses that insight to structure the analysis.
- The feminist art-history argument — which carries a great deal of the paper's interpretive weight — is supported by exactly one reference, Wolfe & Caliskan [5], which is a VLM bias paper, *not* art history. Claiming in §5.2 that a finding "directly aligns with feminist art history literature analyzing the institutional devaluation of craft" while citing only [5] is a misattribution that a humanities readership will notice immediately. Nochlin (1971), Parker's *The Subversive Stitch* (1984), and Pollock (1988) are the obvious omissions.
- The three institutional guidelines in §6.3 are asserted rather than developed, and none is validated — including medium-stratified z-score normalisation, which the authors could trivially apply to their own data and report.

Required: substantially deepen the socio-technical and art-historical engagement, and turn §6.3 from a list into a demonstrated contribution. The collection promises "implementable frameworks based on theoretical insights"; at present the paper supplies neither the theory nor the demonstration.

3. Major Concerns

3.1 The reported effects run in the opposite direction from the paper's interpretation

This is the central problem and it recurs at every level of the manuscript.

(a) Composite scores. Under OpenAI CLIP, female-attributed works score *higher* ($\mu = +0.0107$) than male-attributed works ($\mu = -0.0008$). Under OpenCLIP, female-attributed works score *substantially higher* ($\mu = +0.0896$ vs $+0.0343$). RQ1 asks whether CLIP assigns "significantly lower aesthetic valuation to artworks created by female artists." The answer in both models is unambiguously **no** — and in OpenCLIP the significant effect is in the *opposite* direction. The abstract nonetheless characterises the OpenCLIP result as evidence that "uncurated web training sets ingest persistent demographic valuation disparities," language that any reader will parse as female disadvantage. Figure 5's caption is the only

place in the manuscript that states the direction plainly ("favoring female-attributed works"). The abstract, §6.2, and §7 all omit it.

(b) Prompt Set 3, the paper's most-cited secondary finding. Table 6 reports OpenAI CLIP Set 3 (Influence vs. Craft): Male mean = -0.0303 , Female mean = -0.0127 . Female-attributed works score **higher**. Figure 6 confirms this visually. Yet Appendix A.2 states that Set 3 "yields a statistically significant score penalty for female-attributed works," and the abstract frames the same result as "reflecting gendered historical art taxonomies." The Fig. 6 caption says OpenAI CLIP "exhibits significant devaluation under Prompt Set 3" without saying of whom. **The paper's craft-devaluation narrative is contradicted by its own Table 6.** This must be corrected or withdrawn.

(c) The rank-biserial sign convention is self-contradictory. §4.4 defines r as ranging "from $r = -1$ (complete female dominance) to $r = +1$ (complete male dominance)." For the composite OpenAI CLIP comparison, $U = 867$, $n_M \cdot n_F = 1,904$, so $r = 1 - 2(867)/1904 = +0.0893$ — the reported value, and by the authors' own definition it indicates *male* dominance. But males score lower. The same contradiction holds for OpenCLIP ($r = +0.2973$ with females scoring higher). Meanwhile Table 6 reports *negative* r values for Set 3 where females score higher. The sign convention is applied inconsistently within a single table.

Required action. State the direction of every effect explicitly, in the abstract, every results paragraph, every figure caption, and the conclusion. Fix the rank-biserial convention and apply it consistently. Then — and this is the substantive part — engage seriously with what a female-favouring valuation shift in LAION-2B actually means. There is a real and interesting paper here (possible explanations include era confounding, subject-matter distribution, or alt-text describing women artists' works with different vocabulary), but it is not the paper currently written.

3.2 Century is documented as a confounder and then omitted from the regression

The authors construct century buckets in §3.2.2 and demonstrate in Table 2 that artist gender is strongly associated with era (3.1% female pre-18th century → 34.2% in the 20th century and modern). CLIP is exquisitely sensitive to period style. Era is therefore, on the authors' own evidence, the most obvious confounder in the study — and it does not appear in Equation 9.

This compounds with the sampling frame: Department 11 (European Paintings, 13th–19th c.) and Department 21 (Modern and Contemporary) are near-disjoint in period, and medium composition is largely determined by department. The medium dummies are thus partly proxying for era, which means the paper's central claim ("medium dominance") is not separable from "era dominance" as currently specified.

Required action. Refit both models with century as a covariate. Report whether the medium coefficients survive. If they do not, the paper's thesis changes.

3.3 No correction for multiple comparisons, and the prompt-level findings do not survive one

Six prompt-level tests are reported in Table 6 (3 sets $\times$ 2 models), plus two composite tests. Ordered p-values: 0.0105, 0.0292, 0.0944, 0.1227, 0.3156, 0.4414. Under Holm-Bonferroni at $\alpha = 0.05$, the first comparison is tested against $0.05/6 = 0.0083$; 0.0105 exceeds it, and the procedure stops. **No prompt-level result survives correction.** Under plain Bonferroni the same conclusion holds.

Both prompt-level findings highlighted in the abstract and conclusion (OpenAI Set 3, $p = 0.0292$; OpenCLIP Set 2, $p = 0.0105$) are therefore not defensible as stated. The composite OpenCLIP result ($p = 0.0228$ against two tests) survives Bonferroni only marginally.

Required action. Apply and report a correction. Reframe the prompt-sensitivity section as exploratory, which is in fact its legitimate and still-valuable role — the point that prompt choice can "obscure or artificially fabricate demographic disparities" is well made and does not require any individual comparison to be significant.

3.4 Sample size is unjustified given the available data

The pipeline yields 882 attributed works with resolved gender. The evaluation cohort is 96. No justification is offered for discarding 89% of usable observations. CLIP ViT-B/32 inference over 882 images is a matter of minutes on commodity hardware; there is no compute argument available here.

The consequences are not cosmetic. The authors themselves report power of 0.42 for small effects and repeatedly hedge their central negative finding on this basis ("medium dominance rather than definitive demographic invariance"). The title promises to *disentangle* algorithmic bias from archival artifacts; at $N = 96$ with $1 - \beta = 0.42$, the study is not powered to do so.

More acutely: the sculpture cell contains 8 works, of which 1 is female-attributed. The sculpture coefficient ($B = -0.2873$) — which alone drives most of Model 1's $R^2 = 0.450$ — rests on 8 observations, and the within-sculpture gender contrast is 7 vs 1. This is a positivity/overlap violation, not merely a leverage issue, and reporting max Cook's $D < 0.18$ does not address it.

Required action. Re-run on the full attributed set ($n = 882$, or the subset with retrievable images). This single change resolves the power objection, stabilises the medium coefficients, and permits the gender $\times$ medium interaction the authors say they cannot estimate. If there is a reason the full set cannot be used, state it explicitly.

3.5 Inferential strategy is internally inconsistent

§4.4 rejects normality (Shapiro-Wilk $W = 0.812$, $p < 0.001$) and adopts non-parametric testing on precisely these grounds. §4.5 then specifies OLS with $\varepsilon \sim N(0, \sigma^2)$ and reports classical t-statistics and p-values on the same outcome — an outcome with reported

skewness of -2.368 and kurtosis of 25.166 . The bivariate analysis and the multivariate analysis rest on contradictory distributional premises.

No residual diagnostics are reported. Huber regression is mentioned as a cross-validation but its coefficients are never shown.

Required action. Report HC3 heteroskedasticity-robust or bootstrapped standard errors as the primary inferential basis; report residual diagnostics; report the Huber coefficients in a table rather than asserting agreement; and add a sensitivity analysis excluding sculptures entirely.

3.6 Construct validity of the valuation measure is not established

The paper's dependent variable is a softmax probability differential between phrases such as "an important masterpiece of fine art" and "a minor, forgettable work of art." It is treated throughout as a measure of *artistic valuation* without any external validation. There is no benchmarking against human ratings, exhibition frequency, catalogue-raisonné inclusion, scholarly citation, or any other criterion.

This matters for two reasons. First, for an AI & SOCIETY readership, an unvalidated operationalisation of "value" carrying this much interpretive weight will not pass. Second, the measure is mechanically dependent on the four neutral baseline prompts, which are chosen without justification and never varied.

Required action. Either validate the measure against an external criterion on a subsample, or reframe the outcome throughout as "relative prompt alignment" and drop the language of valuation and aesthetic judgment. Add a sensitivity analysis varying and removing the baseline prompt set.

3.7 The two-model comparison cannot support the causal claim made of it

§1.2 states that the cross-model comparison "prov[es] that pretraining dataset curation directly dictates" the mechanism of bias. OpenAI CLIP and OpenCLIP ViT-B/32 differ in dataset *curation*, dataset *size* (400M vs 2B), training recipe, batch size, schedule, and total compute. Curation is one of at least five differences. With $n = 2$ models and no seed variation, no causal attribution to curation is available.

Required action. Remove "proving" and equivalent language. Either add models that vary curation while holding scale approximately constant (e.g. LAION-400M vs LAION-2B checkpoints, or DataComp variants) — which would make this a genuinely strong contribution — or frame the comparison as a descriptive contrast consistent with a curation hypothesis.

3.8 Gender inference method: validity and ethics

Three distinct problems.

(a) The Met's `artistGender` field. §3.2.1 says the field is queried first and is "unpopulated" in fewer than 8% of records. In the Met Open Access dataset this field is, as a documented quirk, populated only for female artists and left blank otherwise. If the authors treated blanks as non-informative and fell through to name inference, the pipeline is defensible but the description is wrong. If blanks were read as male, the analysis is invalid. **This must be clarified explicitly and unambiguously.**

(b) No validation of the classifier. `gender-guesser` accuracy is never measured on this corpus. The tool is a dictionary lookup derived from Jörg Michael's name list — it is not "trained on international census frequency tables" as stated, and it has known and systematic failure modes on historical, Latinised, and non-Western names, which is exactly this corpus. Manually validate against a labelled random subsample ($n \approx 100$) and report accuracy and error direction.

(c) Ethics of algorithmic gender inference. A paper on responsible AI that imposes binary, name-inferred gender on identifiable historical individuals, in a venue like AI & SOCIETY, must address the ethics of that operation directly. The §3.2.1 paragraph on the 41.2% exclusion is genuinely good and should be kept. But there is no engagement with Keyes (2018) on automated gender recognition, no acknowledgement that the binary is imposed rather than found, and no reflexive note on the study's exclusion of non-Western naming conventions — which means the audit is, in practice, an audit of Euro-American attributions. The Declarations state "Ethics approval: Not applicable"; a data-ethics statement is warranted regardless of IRB status.

4. Numerical and Internal Consistency Errors

These require correction independently of everything above. I checked the reported arithmetic; several figures do not reconcile.

1. **Table 1 vs Table 2 are irreconcilable.** Table 1 reports 618 unknown/unattributed works = 41.20% of the corpus. Table 2 reports "Unknown (%)" per era as 10.5%, 7.3%, 9.0%, and 7.7%. No weighting of four era-level rates in the 7–11% range can produce a corpus-level rate of 41.2%. Either Table 2 is computed on a different denominator (in which case the Unknown column is meaningless as labelled) or one of the two tables is wrong.
2. **Table 1 percentages appear copied from the cohort.** $625/882 = 70.86\%$ and $257/882 = 29.14\%$, not the reported 70.83% and 29.17%. Those latter values are exactly $68/96$ and $28/96$ — the evaluation cohort proportions. This looks like a transposition, and it undermines confidence in the surrounding numbers.
3. **§3.3 nationality figures contradict Figure 2.** The text reports French 28.4%, American 21.2%, Italian 12.6%, British 9.1%, Dutch 7.2%. Figure 2 shows French $\approx 34\%$, German $\approx 14\%$, Netherlandish $\approx 11.5\%$, Spanish $\approx 11.5\%$, Dutch $\approx 7\%$, British $\approx 6\%$, Italian $\approx 5\%$, American $\approx 5\%$. German and Netherlandish do not appear in the text

at all; American is second in the text and eighth in the figure. These cannot both be right.

4. **Figure 2 caption says "top 15 nationalities."** Ten bars are plotted.
 5. **K = 6 does not match the stated prompt set.** §4.2 defines $P = \{p_{\text{high}}^{(j)}, p_{\text{low}}^{(j)}\} \cup P_{\text{base}}$ and asserts $K = 6$. Three prompt pairs give six value prompts; adding four baselines gives ten. If instead each prompt set is normalised separately (2 value + 4 baseline = 6), then Equation 5 averages differentials computed under three *different* softmax normalisations, which is not the same quantity and needs justification. Equations 3–5 must be made consistent with the actual procedure.
 6. **τ is defined twice, incompatibly.** Equation 1 calls it "a learned log-temperature scaling parameter"; §4.3 calls it "the constant scaling multiplier $\tau = 100$." State that the learned logit scale converges to ≈ 100 and is held fixed at inference.
 7. **One Shapiro–Wilk statistic is reported for two distributions.** " $W = 0.812, p < 0.001$ " is said to hold "across both model score distributions." Report separately.
 8. **Appendix A.1 conflates a regression coefficient with a score.** "severe negative softmax probability differentials ($S_{\text{val}} \approx -0.287, p < 0.001$)" quotes the sculpture *coefficient* from Table 7 as if it were an observed score value.
 9. **§6.1's Simpson's paradox claim is incorrect.** The gender coefficient does not reverse sign under adjustment; it attenuates from -0.0114 (raw) to -0.0157 (adjusted) and remains negative. This is ordinary confounding/attenuation, not Simpson's paradox. Either identify an actual reversal or remove the claim and the Simpson citation.
 10. **§4.5 asserts $VIF < 2.5$ for all predictors** but the values are not shown, and with one observation in the female-sculpture cell the relevant diagnostic is overlap, not collinearity.
 11. **Model 2 degrees of freedom are omitted** ($F = 4.964$ with no df reported), and Model 2's $R^2 = 0.251$ is absent from the abstract while Model 1's is featured.
 12. **Bootstrap B = 1,000 is low** for percentile intervals. Use $\geq 10,000$; there is no cost reason not to.
 13. **Post-hoc power is not the right tool.** Observed power computed from observed effects is a monotone transform of the p-value (Hoenig & Heisey 2001). Replace with a minimum detectable effect size analysis at the design stage, which makes the same point defensibly.
 14. **"over 470,000 cultural heritage artifacts"** — verify against the current Met Open Access figure and give an access date consistent with reference [22], which lists "Accessed: 2026-07-24 (2023)."
 15. **Reference [31] attribution.** `(gender-guesser)` is authored by Israel Saeta Pérez, deriving from Jörg Michael's `(gender.c)`; the name list is not census-derived. Correct the attribution and the description.
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5. Presentation

- **Title.** "Museum Collections" is plural; the study uses one museum. And "disentangling" overpromises relative to $1 - \beta = 0.42$. Consider *"A Confound-Controlled Audit of Vision-Language Valuation in a Museum Collection."*
 - **Abstract.** Currently reports Model 1 R^2 but not Model 2's, and omits the direction of every effect. Both must change.
 - **Figure 1** conveys two numbers and duplicates Table 1. Consider removing it, or replacing it with the gender $\times$ medium $\times$ era cross-tabulation that would actually support the confounding argument.
 - **Figure 3** (framework diagram) is useful and well constructed.
 - **Figures 4 and 5** should share y-axis scaling to make the cross-model comparison legible, and should be annotated with the direction of the effect.
 - **Double-blind compliance.** The Declarations section (author contributions, funding) should be removed from the anonymised manuscript file per Springer's double-blind procedure.
 - **Data and code.** "Available upon request" is below the standard now expected. The Met object IDs are public and non-sensitive; deposit the 96 (or 882) object IDs, the enriched metadata table, and the scoring scripts in a citable repository with a DOI. For a paper arguing that institutions should adopt auditing protocols, this is close to a self-consistency requirement.
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6. What Is Genuinely Good Here

I want to be explicit that I think this work is worth persisting with.

- The core methodological argument — that observational VLM audits on archival data are confounded by medium, and that this is measurable rather than merely assertable — is correct, important, and under-served in the literature.
 - The §3.2.1 discussion of the 41.2% exclusion is the strongest passage in the paper. The recognition that name-based filtering systematically removes historically marginalised female labour, and that this is an "uneliminable boundary condition," is exactly the kind of reflexive framing this collection is looking for. Expand it.
 - The prompt-sensitivity design is a real contribution. The demonstration that a single prompt choice can flip conclusions is more valuable than any individual significance test in the paper, and it does not depend on those tests surviving correction.
 - The cross-model design is the right instinct even though the current implementation cannot support the causal claim.
 - Figure 3 and the overall structure are clear and easy to follow.
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7. Checklist of Required Actions

Must fix before this can be reconsidered:

1. State the direction of every reported effect; correct the Prompt Set 3 interpretation in the abstract, §5.2, Fig. 6 caption, §7, and Appendix A.2.
2. Fix the rank-biserial sign convention and apply it consistently.
3. Re-run on the full attributed sample ($n \approx 882$) or justify $N = 96$ explicitly.
4. Add century/era to the OLS specification.
5. Apply and report multiple-comparison correction; reframe prompt-level results as exploratory.
6. Clarify unambiguously how the Met `artistGender` field was handled.
7. Reconcile Table 1 with Table 2; reconcile §3.3 with Figure 2; fix the Table 1 percentages.
8. Resolve the $K = 6$ prompt-set inconsistency and align Equations 3–5 with the actual procedure.
9. Replace classical OLS inference with robust/bootstrapped standard errors; report residual diagnostics and Huber coefficients.
10. Remove causal language regarding curation ("proving," "directly dictates").

Should fix:

11. Validate `gender-guesser` on a labelled subsample; report accuracy.
 12. Add a data-ethics discussion of algorithmic gender inference (Keyes 2018 and related).
 13. Validate the valuation construct externally, or reframe it as prompt alignment.
 14. Add sensitivity analyses: excluding sculpture; varying baseline prompts; gender $\times$ medium interaction.
 15. Demonstrate the proposed medium-stratified z-score normalisation on the paper's own data.
 16. Add substantive feminist art-history sources (Nochlin, Parker, Pollock) and stop citing [5] for art-historical claims.
 17. Replace post-hoc power with minimum-detectable-effect analysis; increase bootstrap resamples to $\geq 10,000$.
 18. Deposit data and code with a DOI.
 19. Deepen the critical-archival-studies engagement so the paper reads as an AI & SOCIETY contribution rather than a CV-venue audit.
 20. Correct reference [31]; verify the Met collection size and access date.
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8. Confidential Comments to the Editors

My concern is not the quality of the empirical machinery, which is competent, but the mismatch between what the numbers show and what the paper says they show. In at least three places — the abstract, §6.2, and Appendix A.2 — the manuscript describes results that favour female-attributed works using language that reads as female disadvantage. Figure 5's caption is the sole place the direction is stated correctly. I do not think this is deliberate; it reads like a paper written to an anticipated conclusion whose data went the other way. But it is a serious problem, and it is the sort of thing that, if published, would be caught quickly and publicly.

The good news is that the underlying study is salvageable and the fix is largely mechanical: state directions honestly, re-run at $n \approx 882$ with era controlled, correct for multiplicity, and rewrite the interpretation around what was actually found. A finding that LAION-2B assigns *higher* valuation scores to female-attributed works after confound adjustment is interesting and publishable in its own right, and would make a better paper than the one submitted.

On collection fit: I would encourage the editors, if they invite a revision, to ask the authors explicitly to strengthen the humanities and archival-studies engagement. The call is interdisciplinary and asks for implementable frameworks grounded in theory. The current §6.3 recommendations are a bulleted list, not a framework, and the authors are well positioned to demonstrate at least the normalisation proposal empirically.

I would be willing to review a revised version.